

Category	Measure	Description
End	F1	Comparison of how often the SIMULATOR ends the conversation whens the human ends it
Lexical	Utterance Length ^s	Log-transformed number of words (for English and Russian) or characters (for Chinese)
	Perplexity ^s	Log-transformed perplexity of the utterance, calculated using ³ . For Russian we use rugpt3small_based_on_gpt2 (Zmitrovich et al., 2023), and for Chinese gpt2-chinese-cluecorpussmall (Zhao et al., 2019).
Syntactic	Part of Speech ^d	Distribution of the utterance’s part of speech tags from spaCy, trained using language-specific models (en_core_web_sm, zh_core_web_sm, ru_core_news_sm).
	Dependency Tree Depth ^s	Log-transformed depth of the dependency tree from spaCy.
	Tree Breadth ^s	Log-transformed number of leaf nodes.
	Tree Dependency Distance ^s	Log-transformed average distance between dependents.
Semantic	SBERT ^v	Cosine similarity of utterance embeddings from the Alibaba-NLP/gte-multilingual-base language model for all three languages (Zhang et al., 2024)
Style	Punctuation ^d	Distribution of punctuation characters
	Sentiment ^s	Distribution of positive, neutral, and negative sentiment using lxyuan/distilbert-base-multilingual-cased-sentiments-student for Chinese and blanchefort/rubert-base-cased-sentiment for Russian
	Toxicity ^s	Toxicity of tone and content, as judged by annotatorss-nlp/russian_toxicity_classifier for Russian and textdetox/xlmr-large-toxicity-classifier

Table 4: Measures used to evaluate how well LLMs capture properties of human responses at Turn 3 of a conversation in Russian and Chinese. Superscript indicates whether the difference between human and SIMULATOR measurements are (s) scalar values (compared with l1-distance), (d) probability distributions (compared with Jensen-Shannon divergence), or (v) vector embeddings (compared with cosine distance).

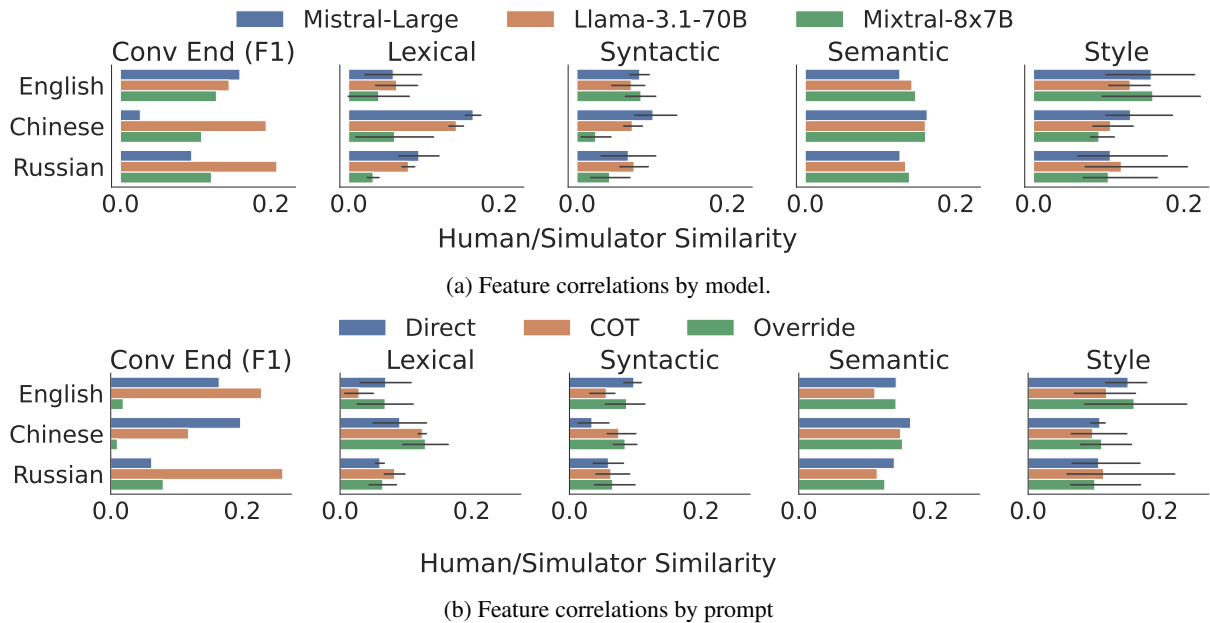


Figure 10: We compare the performance across models and prompts, for individual metrics available in English, Chinese, and Russian. Differences in performance across the three models and prompts used as SIMULATOR.

	Metric	Llama-3.1-8B	Llama-3.1-70B	Llama-3-70B	Mistral-7B	Mixtral-8x7B	Mistral-Large-123B	Phi-3-14B	Qwen2-72B	Command-R-35B	Human
Conv End	F1	0.023	0.030	0.024	0.070	0.118	0.153	0.154	0.145	0.214	0.600
Lexical	Utterance Length	0.089	0.111	0.103	0.075	0.075	0.103	0.103	0.099	0.093	0.168
	Average Word Length	0.115	0.127	0.128	0.124	0.155	0.143	0.096	0.132	0.129	0.150
	Perplexity	0.047	0.077	0.041	0.008	0.027	0.045	0.020	0.034	0.046	0.033
	Typographical Errors	0.187	0.218	0.153	0.152	0.128	0.210	0.160	0.223	0.187	0.248
Syntactic	Part of Speech	0.073	0.087	0.081	0.080	0.079	0.084	0.074	0.085	0.071	0.136
	Dependency Tree Depth	0.106	0.126	0.122	0.087	0.079	0.102	0.099	0.100	0.099	0.161
	Tree Breadth	0.065	0.071	0.067	0.033	0.045	0.072	0.051	0.047	0.052	0.213
	Tree Dependency Distance	0.091	0.106	0.091	0.056	0.054	0.085	0.064	0.072	0.075	0.237
Semantic	SBERT	0.135	0.142	0.140	0.154	0.153	0.138	0.147	0.145	0.130	-0.006
	LIWC	0.073	0.078	0.073	0.087	0.089	0.073	0.080	0.075	0.069	0.083
	Prompt Type	0.282	0.297	0.293	0.295	0.288	0.278	0.279	0.269	0.261	0.190
Style	Punctuation	0.046	0.076	0.072	0.082	0.052	0.068	0.068	0.062	0.061	0.141
	Capitalization	0.074	0.179	0.118	0.103	0.081	0.134	0.060	0.071	0.091	0.551
	Sentiment	0.193	0.176	0.162	0.170	0.165	0.154	0.181	0.148	0.182	0.149
	Politeness	0.100	0.104	0.101	0.162	0.148	0.150	0.154	0.112	0.168	0.191
	Formality	-0.015	0.005	0.003	0.028	0.006	0.012	0.012	-0.008	0.016	-0.043
	Toxicity	0.088	0.080	0.092	0.199	0.252	0.233	0.119	0.042	0.215	0.212
	Readability	0.164	0.167	0.166	0.248	0.244	0.229	0.200	0.198	0.196	0.070
	Subjectivity	0.051	0.059	0.060	0.069	0.058	0.065	0.057	0.076	0.065	0.102
	LUAR	0.044	0.048	0.046	0.048	0.047	0.047	0.047	0.045	0.040	-0.003

Table 5: Correlation between SIMULATOR and HUMAN Turn 3 across models in English.

	Metric	Best Prompt	Worst Prompt	Prompt Override	Prompt Direct	Prompt CoT	Human
Conv End	F1	0.019	0.022	0.072	0.245	0.156	0.600
Lexical	Utterance Length	0.205	0.023	0.148	0.112	0.106	0.168
	Average Word Length	0.309	0.069	0.097	0.106	0.185	0.150
	Perplexity	0.125	0.002	0.073	0.039	0.043	0.033
	Typographical Errors	0.250	0.031	0.251	0.242	0.195	0.248
Syntactic	Part of Speech	0.111	0.041	0.08	0.098	0.073	0.136
	Dependency Tree Depth	0.165	0.049	0.154	0.104	0.099	0.161
	Tree Breadth	0.121	0.022	0.052	0.063	0.028	0.213
	Tree Dependency Distance	0.112	0.032	0.120	0.096	0.090	0.237
Semantic	SBERT	0.146	0.125	0.137	0.139	0.123	-0.006
	LIWC	0.093	0.055	0.066	0.078	0.070	0.083
	Prompt Type	0.320	0.184	0.283	0.296	0.277	0.19
Style	Punctuation	0.115	0.062	0.088	0.082	0.056	0.141
	Capitalization	0.208	0.063	0.141	0.061	0.074	0.551
	Sentiment	0.195	0.221	0.135	0.148	0.125	0.149
	Politeness	0.215	0.096	0.112	0.126	0.140	0.191
	Formality	0.008	-0.007	0.015	-0.003	0.010	-0.043
	Toxicity	0.219	0.095	0.167	0.183	0.117	0.212
	Readability	0.250	0.201	0.205	0.224	0.186	0.070
	Subjectivity	0.089	0.038	0.057	0.071	0.072	0.102
	LUAR	0.055	0.032	0.042	0.043	0.042	-0.003

Table 6: Correlation between SIMULATOR and HUMAN Turn 3 across prompts in English.

	Metric	Mixtral-8x7B	Llama-3.1-70B	Mistral-Large-123B	Prompt Override	Prompt Direct	Prompt CoT
Conv End	F1	0.108	0.194	0.027	0.010	0.197	0.118
Lexical	Utterance Length	0.009	0.133	0.155	0.095	0.050	0.119
	Perplexity	0.112	0.151	0.174	0.163	0.130	0.130
Syntactic	Part of Speech	0.054	0.068	0.081	0.060	0.072	0.070
	Dependency Tree Depth	0.001	0.074	0.073	0.073	0.006	0.053
	Tree Breadth	0.019	0.093	0.152	0.108	0.036	0.113
	Tree Dependency Distance	0.026	0.058	0.098	0.097	0.022	0.064
Semantic	SBERT	0.160	0.160	0.162	0.158	0.170	0.155
Style	Punctuation	0.078	0.062	0.105	0.080	0.095	0.066
	Sentiment	0.107	0.132	0.185	0.157	0.117	0.150
	Toxicity	0.076	0.113	0.096	0.098	0.115	0.079

Table 7: Correlation between SIMULATOR and HUMAN Turn 3 across models and prompts in Chinese.

	Metric	Mixtral-8x7B	Llama-3.1-70B	Mistral-Large-123B	Prompt Override	Prompt Direct	Prompt CoT
Conv End	F1	0.121	0.208	0.095	0.080	0.062	0.262
Lexical	Utterance Length	0.040	0.088	0.120	0.086	0.067	0.099
	Perplexity	0.024	0.071	0.066	0.044	0.054	0.067
Syntactic	Part of Speech	0.050	0.075	0.081	0.063	0.073	0.063
	Dependency Tree Depth	0.087	0.101	0.128	0.118	0.091	0.108
	Tree Breadth	0.020	0.076	0.035	0.046	0.039	0.043
	Tree Dependency Distance	0.016	0.051	0.029	0.036	0.034	0.037
Semantic	SBERT	0.138	0.133	0.126	0.131	0.145	0.119
Style	Punctuation	0.069	0.077	0.069	0.068	0.085	0.062
	Sentiment	0.065	0.069	0.059	0.065	0.067	0.059
	Toxicity	0.164	0.205	0.178	0.171	0.170	0.223

Table 8: Correlation between SIMULATOR and HUMAN Turn 3 across models and prompts in Russian.

	Lexical	Semantic	Style	Syntactic	Overall
ai_turn_2_capitalization	-0.25***	-0.33***	-0.06	-0.16***	-0.20***
ai_turn_2_log_word_count	0.00	0.03***	0.06***	0.01***	0.02***
ai_turn_2_politeness	-0.14***	-0.03***	-0.09***	0.05***	-0.05***
ai_turn_2_sentiment	0.03***	0.07***	0.06***	0.00	0.04***
ai_turn_2_subjectivity	-0.02	-0.05***	0.06***	-0.09***	-0.02***
ai_turn_2_toxicity	-0.05***	0.01	-0.02	-0.07***	-0.03**
ai_turn_2_typo	-0.13***	-0.10***	-0.12***	-0.02*	-0.09***
ai_turn_2_word_length	-0.14***	-0.05***	0.08***	0.02***	-0.02***
const	0.52***	0.18***	-0.43***	-0.05***	0.06***
human_turn_1_capitalization	0.06***	-0.05	-0.16***	-0.41***	-0.14***
human_turn_1_log_word_count	0.03***	0.03***	0.06***	0.02***	0.03***
human_turn_1_politeness	0.22***	0.20***	0.09***	0.16***	0.17***
human_turn_1_sentiment	-0.17***	-0.24***	-0.10***	0.03***	-0.12***
human_turn_1_subjectivity	0.10***	0.11***	0.06***	0.03***	0.08***
human_turn_1_toxicity	-0.01	-0.01	-0.06***	-0.16***	-0.06***
human_turn_1_typo	-0.17***	-0.19***	-0.15***	-0.06***	-0.14***
human_turn_1_word_length	-0.15***	-0.10***	0.05***	-0.03***	-0.06***
SIMULATOR_Mistral-Large-Instruct	0.04***	-0.01	-0.04***	-0.06***	-0.02***
SIMULATOR_Mixtral-8x7B	-0.03***	-0.01	0.03***	-0.03***	-0.01***
CHATBOT_gpt-3.5-turbo-0125	-0.03**	-0.02	-0.03***	-0.01	-0.02**
CHATBOT_gpt-3.5-turbo-0613	0.04***	0.07***	0.01***	0.02***	0.04***
CHATBOT_gpt-4-0125-preview	-0.03***	-0.01	-0.08***	-0.02***	-0.03***
CHATBOT_gpt-4-0314	0.05***	0.08***	-0.01	0.04***	0.04***
CHATBOT_gpt-4-1106-preview	-0.04***	0.01	-0.07***	-0.02***	-0.03***
Prompt_15	0.04***	0.01	-0.01	0.04***	0.02***
Prompt_19	0.11***	-0.01	-0.15***	0.06***	0.00
subregion_Central Asia	-0.12***	-0.05	-0.06	-0.03	-0.06***
subregion_E Asia	-0.19***	-0.21***	-0.06***	0.06***	-0.10***
subregion_E Europe	-0.10***	-0.01	-0.04***	-0.01**	-0.04***
subregion_Latin America	0.03***	0.05***	0.02	-0.03***	0.02
subregion_N Africa	-0.12***	-0.10***	-0.03***	0.01*	-0.06***
subregion_N America	-0.05***	-0.09***	-0.03*	0.01	-0.04***
subregion_N Europe	0.01	0.03***	0.01	-0.02***	0.01
subregion_Oceania	-0.05***	-0.01	-0.05***	-0.03***	-0.04***
subregion_S Asia	-0.13***	-0.16***	-0.07***	0.01	-0.09***
subregion_S Europe	-0.05***	-0.04***	-0.03**	-0.03***	-0.04***
subregion_SE Asia	-0.19***	-0.15***	-0.08***	0.00	-0.11***
subregion_Sub-Saharan Africa	-0.03	-0.05**	0.03	0.03***	0.00
subregion_W Asia	-0.12***	-0.12***	-0.06***	-0.03***	-0.08***
subregion_W Europe	-0.08***	-0.05***	-0.02**	-0.02***	-0.04***
Observations	296120.00	296122.00	296122.00	296122.00	296122.00
R ²	0.11	0.07	0.16	0.10	0.12
Adjusted R ²	0.11	0.07	0.16	0.10	0.12
Residual Std. Error	0.60	0.77	0.64	0.36	0.46
F Statistic	399.06***	241.83***	628.70***	354.59***	452.18***

Note:

*p<0.05; **p<0.01; ***p<0.001

Table 9: Coefficients for all regressions in §7. Stars represent p-values adjusted for multiple comparisons using a Bonferroni correction (* p<0.05, ** p<0.01, *** p<0.001).